

SQL EXPLORATION QUERIES

Kenya Mobile Money & Agent Network Analytics

These 10 queries will help in understanding the data deeply before building Power BI dashboards.

Purpose: Explore patterns, validate assumptions, identify outliers, and prepare for KPI design.

How to Run These Queries

Option 1: In Command Prompt (Windows)

```
mysql -u root -p -e "USE kenya_mobile_money; [PASTE QUERY HERE]"
```

Option 2: In MySQL Workbench

1. Open MySQL Workbench
2. Connect to kenya_mobile_money database
3. Open a new query tab
4. Paste the query
5. Click the lightning bolt (Execute)

Option 3: In VS Code Terminal

```
mysql -u root -p
```

```
# Password: [Database Password]
```

```
# Then type: USE kenya_mobile_money;
```

```
# Then paste the query
```

THE 10 QUERIES

Query 1: Overall Financial Summary

Purpose: Understand the total scale and profitability of your data

```
SELECT
    COUNT(*) AS total_transactions,
    COUNT(DISTINCT AgentID) AS unique_agents,
    COUNT(DISTINCT CustomerID) AS unique_customers,
    ROUND(SUM(TransactionAmount), 2) AS total_transaction_value,
```

```

ROUND(AVG(TransactionAmount), 2) AS avg_transaction_amount,
ROUND(SUM(Commission), 2) AS total_commission,
ROUND(SUM(ProcessingCost), 2) AS total_processing_cost,
ROUND(SUM(Commission) - SUM(ProcessingCost), 2) AS net_revenue,
ROUND((SUM(Commission) - SUM(ProcessingCost)) / SUM(Commission) * 100, 2) AS net_margin_pct
FROM FactTransactions
WHERE TransactionStatus = 'Success';

```

What to Look For:

- Is net margin positive?
- Average transaction value realistic.
- How many unique agents/customers?

Query 2: Top 10 Agents by Net Revenue

Purpose: Identify star performers and understand revenue concentration

```

SELECT
  a.AgentID,
  a.AgentName,
  l.County,
  a.AgentType,
  COUNT(f.TransactionID) AS transaction_count,
  ROUND(SUM(f.TransactionAmount), 2) AS total_value,
  ROUND(SUM(f.Commission), 2) AS total_commission,
  ROUND(SUM(f.Commission) - SUM(f.ProcessingCost), 2) AS net_revenue,
  ROUND(AVG(f.TransactionAmount), 2) AS avg_transaction_value
FROM FactTransactions f
JOIN DimAgent a ON f.AgentID = a.AgentID
JOIN DimLocation l ON a.LocationID = l.LocationID
WHERE f.TransactionStatus = 'Success'
GROUP BY a.AgentID, a.AgentName, l.County, a.AgentType
ORDER BY net_revenue DESC
LIMIT 10;

```

What to Look For:

- Are top agents concentrated in specific locations?
 - Do Super Agents outperform Standard Agents?
 - Is revenue concentrated in a few agents or spread? (Gini coefficient story)
-

Query 3: Revenue by County (Top 15)

Purpose: Understand geographic performance and identify underperforming regions

```
SELECT
    l.County,
    l.Region,
    COUNT(DISTINCT a.AgentID) AS agent_count,
    COUNT(f.TransactionID) AS transaction_count,
    ROUND(SUM(f.TransactionAmount), 2) AS total_value,
    ROUND(SUM(f.Commission) - SUM(f.ProcessingCost), 2) AS net_revenue,
    ROUND(AVG(f.Commission) - AVG(f.ProcessingCost), 2) AS avg_net_per_transaction,
    ROUND((SUM(f.Commission) - SUM(f.ProcessingCost)) / SUM(f.Commission) * 100, 2) AS margin_pct
FROM FactTransactions f
JOIN DimAgent a ON f.AgentID = a.AgentID
JOIN DimLocation l ON a.LocationID = l.LocationID
WHERE f.TransactionStatus = 'Success'
GROUP BY l.County, l.Region
ORDER BY net_revenue DESC
LIMIT 15;
```

What to Look For:

- Which counties are revenue leaders?
 - Are margins consistent across regions? (Where is it harder to be profitable?)
 - Agent density vs. revenue (e.g., does more agents = more revenue?)
-

Query 4: Revenue per Active Agent (By County)

Purpose: This is a KEY METRIC for your Phase 1 of this project. KPI, identifies agent productivity

```
SELECT
    l.County,
    l.Region,
    COUNT(DISTINCT a.AgentID) AS agent_count,
    COUNT(f.TransactionID) AS total_transactions,
    ROUND(SUM(f.Commission) - SUM(f.ProcessingCost), 2) AS county_net_revenue,
    ROUND((SUM(f.Commission) - SUM(f.ProcessingCost)) / COUNT(DISTINCT a.AgentID), 2) AS
revenue_per_agent,
    ROUND(COUNT(f.TransactionID) / COUNT(DISTINCT a.AgentID), 2) AS transactions_per_agent,
    ROUND(AVG(f.TransactionAmount), 2) AS avg_transaction_value
FROM FactTransactions f
JOIN DimAgent a ON f.AgentID = a.AgentID
JOIN DimLocation l ON a.LocationID = l.LocationID
WHERE f.TransactionStatus = 'Success'
GROUP BY l.County, l.Region
ORDER BY revenue_per_agent DESC;
```

What to Look For:

- **KEY INSIGHT:** Is Nairobi higher or lower than other counties?
- Is revenue per agent growing or shrinking as agent count grows?
- This directly answers the Phase 1 question: "Does agent growth translate to profitability growth?"

Query 5: Provider Performance Comparison

Purpose: Understand how providers differ in volume, value, and profitability

```
SELECT
    p.ProviderName,
    p.ProviderCategory,
    COUNT(f.TransactionID) AS transaction_count,
    ROUND(COUNT(f.TransactionID) * 100 / (SELECT COUNT(*) FROM FactTransactions), 2) AS pct_of_total,
    ROUND(SUM(f.TransactionAmount), 2) AS total_value,
    ROUND(AVG(f.TransactionAmount), 2) AS avg_transaction_value,
```

```

ROUND(SUM(f.Commission), 2) AS total_commission,
ROUND(SUM(f.ProcessingCost), 2) AS total_processing_cost,
ROUND(SUM(f.Commission) - SUM(f.ProcessingCost), 2) AS net_revenue,
ROUND((SUM(f.Commission) - SUM(f.ProcessingCost)) / SUM(f.Commission) * 100, 2) AS margin_pct,
ROUND(SUM(f.TransactionStatus = 'Success') * 100 / COUNT(*), 2) AS success_rate_pct
FROM FactTransactions f
JOIN DimProvider p ON f.ProviderID = p.ProviderID
GROUP BY p.ProviderID, p.ProviderName, p.ProviderCategory
ORDER BY net_revenue DESC;

```

What to Look For:

- Does M-Pesa have highest volume AND profitability?
- Do Airtel and Bank Agency have different margin profiles?
- Success rates; do some providers fail more often?

Query 6: Transaction Type Profitability

Purpose: Which transaction types are actually profitable? (Not just high-volume)

```

SELECT
    t.TransactionType,
    t.Category,
    COUNT(f.TransactionID) AS transaction_count,
    ROUND(COUNT(f.TransactionID) * 100 / (SELECT COUNT(*) FROM FactTransactions), 2) AS pct_of_total,
    ROUND(AVG(f.TransactionAmount), 2) AS avg_amount,
    ROUND(AVG(f.Commission), 2) AS avg_commission,
    ROUND(AVG(f.ProcessingCost), 2) AS avg_processing_cost,
    ROUND(AVG(f.Commission) - AVG(f.ProcessingCost), 2) AS avg_net_revenue,
    ROUND((SUM(f.Commission) - SUM(f.ProcessingCost)) / SUM(f.Commission) * 100, 2) AS margin_pct,
    ROUND(SUM(f.TransactionStatus = 'Success') * 100 / COUNT(*), 2) AS success_rate_pct
FROM FactTransactions f
JOIN DimTransactionType t ON f.TransactionTypeID = t.TransactionTypeID
GROUP BY t.TransactionTypeID, t.TransactionType, t.Category
ORDER BY avg_net_revenue DESC;

```

What to Look For:

- **KEY INSIGHT:** High-volume transaction types might not be the most profitable.
- Which transaction types fail most often?
- Are there quick wins? (High margin, low failure rate)

Query 7: Customer Segment Behavior

Purpose: Understand how different customer segments use the platform

SELECT

```
c.CustomerSegment,
c.AgeGroup,
COUNT(f.TransactionID) AS transaction_count,
COUNT(DISTINCT f.CustomerID) AS unique_customers,
ROUND(COUNT(f.TransactionID) / COUNT(DISTINCT f.CustomerID), 2) AS transactions_per_customer,
ROUND(AVG(f.TransactionAmount), 2) AS avg_transaction_amount,
ROUND(SUM(f.TransactionAmount), 2) AS total_value,
ROUND(AVG(f.Commission) - AVG(f.ProcessingCost), 2) AS avg_net_revenue_per_txn,
ROUND(SUM(f.TransactionStatus = 'Success') * 100 / COUNT(*), 2) AS success_rate_pct
```

FROM FactTransactions f

JOIN DimCustomer c ON f.CustomerID = c.CustomerID

WHERE f.TransactionStatus = 'Success'

GROUP BY c.CustomerSegment, c.AgeGroup

ORDER BY total_value DESC;

What to Look For:

- Which segments have highest transaction frequency?
- Do younger or older customers use it differently?
- Are Business customers more valuable than Mass Market?

Query 8: Transaction Volume by Month (Trend Analysis)

Purpose: Identify seasonal patterns and trends throughout the year

SELECT

```
YEAR(d.DateKey) AS year,
MONTH(d.DateKey) AS month,
d.MonthName,
```

```

COUNT(f.TransactionID) AS transaction_count,
ROUND(SUM(f.TransactionAmount), 2) AS total_value,
ROUND(SUM(f.Commission) - SUM(f.ProcessingCost), 2) AS net_revenue,
ROUND(SUM(f.TransactionStatus = 'Success') * 100 / COUNT(*), 2) AS success_rate_pct
FROM FactTransactions f
JOIN DimDate d ON f.DateKey = d.DateKey
WHERE YEAR(d.DateKey) = 2025
GROUP BY YEAR(d.DateKey), MONTH(d.DateKey), d.MonthName
ORDER BY MONTH(d.DateKey);

```

What to Look For:

- Are there seasonal peaks or troughs?
- Does success rate vary by month?
- Is growth linear or uneven?

Query 9: Failure & Reversal Analysis

Purpose: Understand operational risk and where transactions are failing

```

SELECT
    f.TransactionStatus,
    COUNT(*) AS count,
    ROUND(COUNT(*) * 100 / (SELECT COUNT(*) FROM FactTransactions), 2) AS pct_of_total,
    ROUND(AVG(f.TransactionAmount), 2) AS avg_amount,
    ROUND(SUM(f.TransactionAmount), 2) AS total_value
FROM FactTransactions f
GROUP BY f.TransactionStatus
ORDER BY count DESC;

```

-- By Transaction Type

```

SELECT
    t.TransactionType,
    f.TransactionStatus,
    COUNT(*) AS count,
    ROUND(COUNT(*) * 100 / SUM(COUNT(*)) OVER (PARTITION BY t.TransactionTypeID), 2) AS pct_of_type
FROM FactTransactions f

```

```
JOIN DimTransactionType t ON f.TransactionTypeID = t.TransactionTypeID
GROUP BY t.TransactionType, f.TransactionStatus
ORDER BY t.TransactionType, count DESC;
```

```
-- By Provider
```

```
SELECT
    p.ProviderName,
    f.TransactionStatus,
    COUNT(*) AS count,
    ROUND(COUNT(*) * 100 / SUM(COUNT(*)) OVER (PARTITION BY p.ProviderID), 2) AS pct_of_provider
FROM FactTransactions f
JOIN DimProvider p ON f.ProviderID = p.ProviderID
GROUP BY p.ProviderName, f.TransactionStatus
ORDER BY p.ProviderName, count DESC;
```

What to Look For:

- Overall success rate realistic?
- Which transaction types fail most?
- Do some providers have worse success rates?

Query 10: High-Value Transaction Analysis

Purpose: Understand the distribution of large transactions and outliers

```
SELECT
    -- Quantile breakdown
    ROUND(PERCENTILE_CONT(0.25) WITHIN GROUP (ORDER BY TransactionAmount) OVER (), 2) AS q1_25th_percentile,
    ROUND(PERCENTILE_CONT(0.50) WITHIN GROUP (ORDER BY TransactionAmount) OVER (), 2) AS q2_median,
    ROUND(PERCENTILE_CONT(0.75) WITHIN GROUP (ORDER BY TransactionAmount) OVER (), 2) AS q3_75th_percentile,
    ROUND(PERCENTILE_CONT(0.90) WITHIN GROUP (ORDER BY TransactionAmount) OVER (), 2) AS q4_90th_percentile,
    ROUND(PERCENTILE_CONT(0.95) WITHIN GROUP (ORDER BY TransactionAmount) OVER (), 2) AS q5_95th_percentile,
    ROUND(PERCENTILE_CONT(0.99) WITHIN GROUP (ORDER BY TransactionAmount) OVER (), 2) AS q6_99th_percentile,
```



```

MIN(TransactionAmount) AS min_amount,
MAX(TransactionAmount) AS max_amount
FROM FactTransactions
LIMIT 1;

```

```
-- High-value transactions (top 5%)
```

```

SELECT
    f.TransactionID,
    d.DateKey,
    a.AgentName,
    l.County,
    c.CustomerSegment,
    p.ProviderName,
    t.TransactionType,
    f.TransactionAmount,
    f.Commission,
    f.ProcessingCost,
    f.TransactionStatus
FROM FactTransactions f
JOIN DimAgent a ON f.AgentID = a.AgentID
JOIN DimLocation l ON a.LocationID = l.LocationID
JOIN DimCustomer c ON f.CustomerID = c.CustomerID
JOIN DimProvider p ON f.ProviderID = p.ProviderID
JOIN DimTransactionType t ON f.TransactionTypeID = t.TransactionTypeID
WHERE f.TransactionAmount > (
    SELECT PERCENTILE_CONT(0.95) WITHIN GROUP (ORDER BY TransactionAmount)
    FROM FactTransactions
)
ORDER BY f.TransactionAmount DESC
LIMIT 20;

```

What to Look For:

- Distribution makes sense?
- High-value transactions mostly successful?
- Which agents/locations handle large amounts?

- Are certain transaction types systematically high-value?

Summary: What These Queries Reveal

Query	Reveals	For Power BI Use
1	Overall profitability & scale	Validate data quality
2	Agent performance concentration	Agent detail page / top performers
3	Geographic revenue patterns	Geographic Intelligence page (main map)
4	Agent productivity by location	Revenue per Agent metric (Phase 1 KPI)
5	Provider competitiveness	Provider slicer / comparison visuals
6	Transaction type profitability	Hidden insight for recommendations
7	Customer segment behavior	Customer segment filter / analysis
8	Seasonal trends	Trend line over time
9	Operational risk	Success rate KPI / alert threshold
10	Outlier patterns	Fraud detection foundation